Study on Time Series Data Mining and Trend Forecasting of Hog Price in China

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Abstract

This report takes the monthly average hog price in 22 provinces and cities in China from July 2017 to 2022 as the basis for short-term forecasts of the monthly data for the subsequent 12 months using the SARIMA model in time series analysis in response to the seasonal changes in the data. The results show that the short-term forecast based on the SARIMA model is excellent.

Keywords: Hog price; Time series analysis; SARIMA; Short-term forecasting

1. Introduction

Pork plays a major role in China's dietary structure, while hogs are China's main livestock object and one of the important bulk-traded agricultural products. Pig production capacity is affected by epidemics, feed costs, environmental protection policies and other impacts arising from changes in supply and demand and the breeding process needs to undergo a long breeding time, which leads to substantial fluctuations in hog prices. Huge price fluctuations will not only cause direct damage to the interests of producers and consumers but also have an impact on the national economy. At the same time, pig farming is exposed to changes in temperature and wind (ventilation conditions, etc.) with seasonal changes, which in turn may generate situations such as pup survival and breeding of epidemics. Through the data mining of historical hog prices, we can train and screen models with high accuracy and stable performance to effectively predict the trend of hog prices, which is of great

significance for the scientific guidance of the pig farming industry, and control the price of pork to stabilise the supply and demand of the hog market.

2. Problem & Solution

In recent years, many scholars have carried out forecasting studies on hog prices, such as vector autoregression[1], support vector machine[2], BP neural network[3], grey neural network[4], time series[5], structural equation modelling[6], and combination forecasting method[7] of the above methods, and other methods to forecast hog prices or pork prices. As for the seasonality of hog prices, most of the literature is based on qualitative research and quantitative research is very lacking, while the SARIMA model in time series will probably have a good performance. In this report, the SARIMA model will be constructed based on time series for short-term forecasting of Chinese hog prices.

(1) SARIMA Model

SARIMA model is a seasonal derivative model based on the ARIMA model, which is a famous time series forecasting method proposed by American statisticians Jenkins and Box in 1976, while SARIMA model takes into account the characteristics of trend and seasonality, and it can better express the price fluctuations of agricultural products that are greatly affected by seasonal changes. In recent years, many scholars have made use of the SARIMA model to forecast the prices of agricultural products. For example, Lu et al.[8] used the SARIMA model to predict the market price of navel oranges in China, and the trend coincided with the actual situation; Zheng et al.[9] used the SARIMA model to better analyse the price of vegetables in Shanghai.

ARIMA (p,d,q) is referred to as the autoregressive integrated moving average model[10]. AR represents the autoregressive process, with p being its autoregressive term; MA represents the moving average process, with q being its moving average term; I represents the number of single integer orders, and d represents the number of differences (orders) made when this time series reaches a steady state. SARIMA(p,d,q)(P,D,Q)s model is referred to as the seasonal auto-regressive integrated moving average model, while P, D, and Q represent the parameters of the

seasonal component, and s denotes the seasonal period.

(2) Data Source

The data for this study is derived from the average monthly hog price in 22 provinces and cities in China from July 2017 to June 2023. The initial data (unit: yuan/kg) was obtained from iFinD to commence the analysis.

(3) Original Time Series Analysis

After reading the initial data using Python and pre-processing the data， the monthly average hog price was visualized, as shown in Figure 1. From July 2017 to June 2023, the monthly average price exhibited significant fluctuation, displaying a "down-up-down-up-down-up" trend. The average monthly price fluctuation increased, indicating a "down-up-down" pattern, roughly resembling a double “∧ ” curve. This initial assessment suggests that the time series of the monthly average hog price in China is a non-stationary time series, and it may be necessary to differentiate the series to meet the fundamental conditions for constructing the ARIMA model.

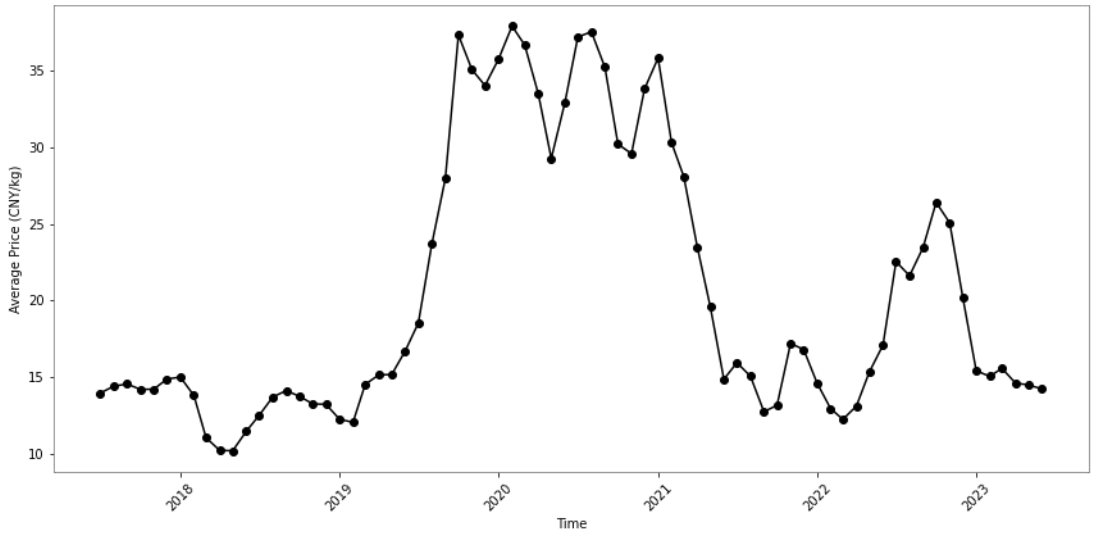


Figure 1 Monthly average hog price in 22 provinces and cities of China, 07/2017-06/2023

(4) First Difference Analysis of the Original Time Series Analysis

The ADF test was used to perform the unit root test on the original series using Python, and its results show that the original series is an unstable time series and needs to be first differenced. The ADF statistics of the first difference of the original

series are smaller than the critical value of its t-statistics at 1%, 5% and 10% test levels, and the p-value is less than 0.01, indicating that the first difference of the original series is a smooth time series (Table 1).The results of the ADF test show that the series of the average price of hogs is a first-order single-integer series, i.e., the ARIMA(p,d,q)model has a d value of 1.

Table 1 Stability test of monthly average hog price series in 22 provinces and cities in China

Variable	ADF Statistic	p-value	Critical Value (1%)	Critical Value (5%)	Critical Value (10%)	Result
Hog Price	-1.939	0.3138	-3.527	-2.904	-2.589	Non-stationary
(D)Hog Price	-5.517	1.9192E-06	-3.529	-2.904	-2.590	Stationary

Using Python to differentiate the original sequence and generate the first differential time series plot, autocorrelation plot, and partial correlation plot. From Figure 2, it is evident that the time series plot, after the first difference processing, is more evenly distributed on both sides of 0, aligning closely with the steady state(stationary).

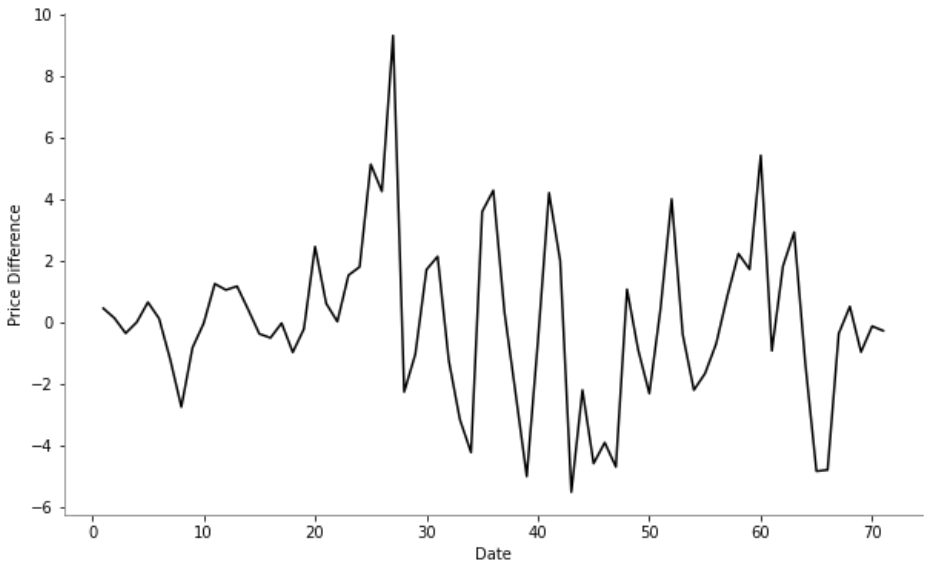


Figure 2 First difference of monthly average hog price

Figure 3 illustrates that the autocorrelation function tends towards 0 after the 1st order, and the subsequent values exhibit sinusoidal vibration, indicating potential seasonal components in the data. The parameter q can take the value of 0 or 1 due to

the significant tendency of the value towards 0 after the 1st order.

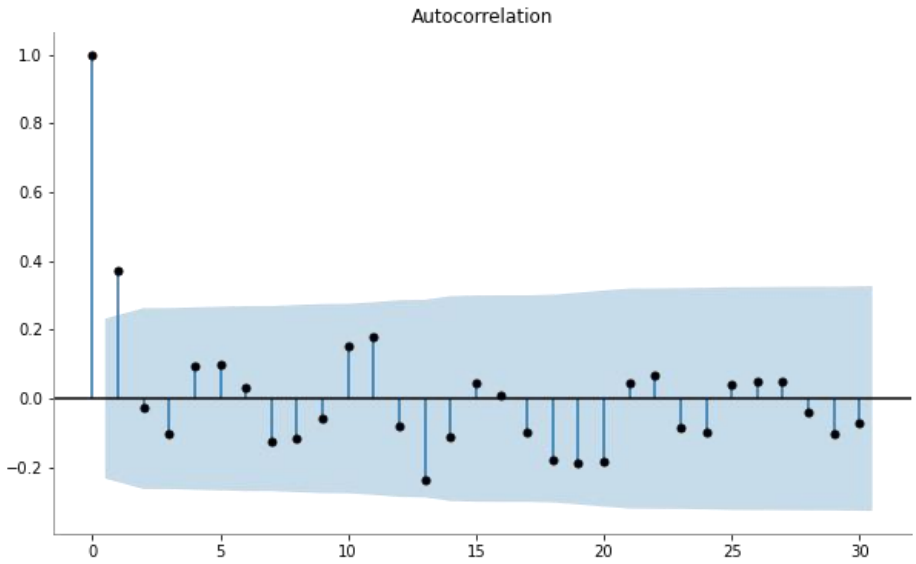


Figure 3 First difference autocorrelation of monthly average hog price

Figure 4 demonstrates that the partial autocorrelation function also tends towards 0 after the 1st order, but it is significant at the 12th, 26th, and 29th orders, possibly influenced by seasonal factors. The parameter p may take the value of 0 or 1.

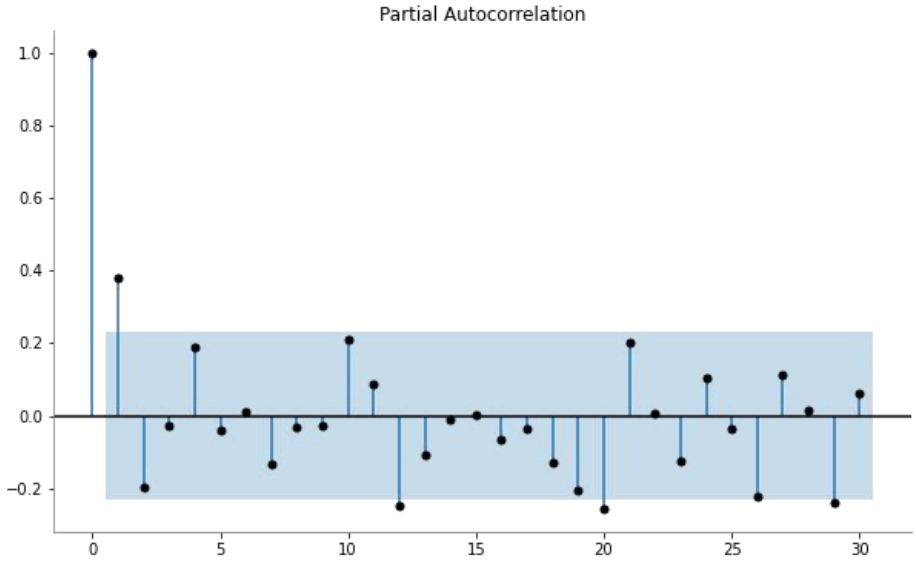


Figure 4 First difference partial autocorrelation of monthly average hog price

Since both the autocorrelation ACF test and the partial autocorrelation PACF test show the possibility of seasonal effects on the series, the SARIMA(p,d,q)(P,D,Q)s

model can be considered and the underlying ARIMA model can be used as a control. Since the effect of seasonal variations on actual hog prices is complex, the seasonal cycle s is considered to be general, i.e. it may take values of 4 or 12.

(5) Model Parameter Estimation and Test

Exploring all possible combinations of p, d, and q, the model is trained on the monthly average hog price from 22 provinces and cities in China, spanning July 2017 to June 2022. It then forecasts the average prices from July 2022 to June 2023. The optimal model is selected based on the Akaike Information Criterion (AIC), which balances the likelihood function against the number of parameters; a lower AIC indicates a superior model fit. According to Table 1, the SARIMA(1,1,0)(1,1,0,12) configuration outperforms other combinations.

Table 2 Different combinations of model tests

Model Category	Model	Seasonal Order	AIC	BIC
Non-Seasonal	(0, 1, 0)	(0, 0, 0, 0)	281.683	283.743
ARIMA	(1, 1, 0)	(0, 0, 0, 0)	275.017	279.138
	(0, 1, 1)	(0, 0, 0, 0)	270.176	274.262
	(1, 1, 1)	(0, 0, 0, 0)	271.786	277.915
Seasonal	(1, 1, 1)	(1, 1, 1, 12)	1356.036	1363.518
s=12	(1, 1, 0)	(1, 1, 0, 12)	193.241	197.820
SARIMA	(0, 1, 1)	(0, 1, 1, 12)	1056.383	1060.872
	(1, 1, 1)	(0, 1, 1, 12)	1355.656	1361.642
Seasonal	(1, 1, 1)	(1, 1, 1, 4)	252.849	262.308
s=4	(1, 1, 0)	(1, 1, 0, 4)	266.750	272.487
SARIMA	(0, 1, 1)	(0, 1, 1, 4)	248.774	254.449
	(1, 1, 1)	(0, 1, 1, 4)	250.405	257.972

3. Performance Evaluation&Comparison

Utilizing the monthly average hog price from 22 provinces and cities in China spanning July 2017 to June 2022, the seasonal SARIMA(1,1,0)(1,1,0,12) model is

trained to predict the monthly average price one year ahead. The prediction results are presented in Figure 5 and Table 3, demonstrating a similar trend between the predicted and observed values, showing a pattern of decline, followed by an increase, and then another decline. However, the peak of the predicted value lags by approximately 2 months and is generally lower than the actual value. The average error between the predicted and actual values is 3.37 yuan/kg, with an average deviation of 16.92%.

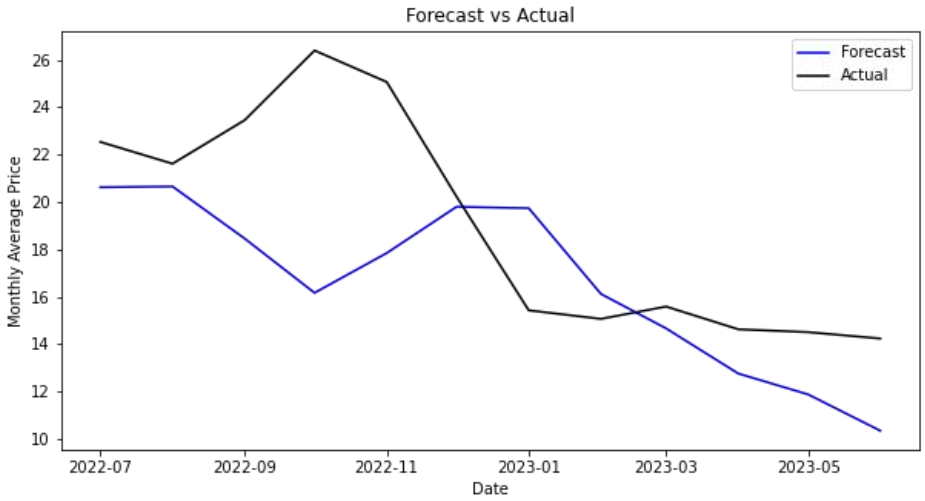


Figure 5 Comparison of forecast and actual values

Table 3 Comparative table of forecast and actual values

Date	Forecast Value	Actual Value	Absolute Error (¥/kg)	Relative Error (%)
2022-07	20.63	22.54	1.91	8.49
2022-08	20.66	21.62	0.96	4.44
2022-09	18.46	23.46	5.00	21.32
2022-10	16.17	26.4	10.23	38.76
2022-11	17.85	25.07	7.22	28.81
2022-12	19.80	20.23	0.43	2.11
2023-01	19.74	15.43	4.31	27.94
2023-02	16.12	15.07	1.05	6.99
2023-03	14.67	15.59	0.92	5.92

2023-04	12.76	14.63	1.87	12.77
2023-05	11.88	14.51	2.63	18.12
2023-06	10.35	14.24	3.89	27.33

4. Conclusion & Discussion

Upon analysis, the error from September 2022 to January 2023 is substantial, while errors in other months remained within an acceptable range. A review of pertinent literature allowed us to dissect the transient surge in the hog price in late 2022, attributable to the industry's significant losses earlier that year, leading to a new cycle of capacity reduction. The nadir of the hog price in May 2022, the upward movement of the feed cost, coupled with rising feed costs and the industry's caution regarding long-term expectations of price increases, has led to a slow recovery of capacity during periods of price increases, prompting a more robust rebound in the hog price from July to November. However, by November, as feed prices decreased and pig production capacity largely recovered, combined with weakening demand for stocking near the Spring Festival and stabilizing policy interventions, the hog price began to recede. To sum up, the interplay of supply dynamics, farming costs, policy measures, and farmer expectations collectively influenced hog price operations in the latter half of 2022.

Given the current model's limitations in assimilating the latest data to reflect market shifts and considering the potential impact of various external factors, dynamic model updating and the incorporation of external variables(e.g. SARIMAX, where X stands for exogenous variable) could enhance predictive accuracy. Despite the volatility induced by extreme supply relations and other factors in the second half of 2022, the SARIMA model's forecasts maintained a low average error rate and captured trends that closely mirrored actuality, affirming the model's relative reliability. This research corroborates the viability of employing the SARIMA model for short-term hog price forecasting.

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